

Predicting bycatch of endangered marine species with a hurdle mixed model

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joint work with J. Mills Flemming (Dalhousie) and A. H. Welsh (Australian National University)

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Research Center for Statistics

International Biometric Conference, 2018, Barcelona

Motivation

Bycatch Reduction Engineering Program – 2017 Awards :: Office of Sustainable Fisheries - Mozilla Firefox

Google Calendar - Web B Bycatch Reduction Eng... +

www.nmfs.noaa.gov/sfa/publications/feature_stories/2017/brep-2017-awards.html

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2017 Award Recipients

APPLICANT	PROJECT TITLE
Pfizer Institute of Environmental Research	Developing radio and satellite smart buoys to reduce bycatch.
Gettysburg College	Developing and testing a multi-sensory bycatch reduction strategy to reduce sea turtle bycatch in gillnet and pound net fisheries
University of Hawaii	1) Assessing the occurrence and severity of decompression sickness in marine turtles incidentally captured by trawl fisheries. 2) Habitat use, movement behavior, and residency of oceanic whitetip sharks, <i>Carcharhinus longimanus</i> found in association with fish aggregating devices in Hawaii: Identifying strategies to reduce mortality of a threatened species.
Professor Evan Evenden	1) Turtles Sublethally and Chronically Injured as Threat and Risk

Bycatch Reduction Engineering Program – 2017 Awards
August 29, 2017

NOAA Fisheries has awarded more than \$2.3 million to partners around the country to support innovative bycatch reduction research projects through its Bycatch Reduction Engineering Program. Bycatch of various species—fish, marine mammals, or turtles—can have significant biological, economic, and social impacts. Preventing and reducing bycatch is a shared goal of fisheries managers, the fishing industry, and the environmental community.

Working side-by-side with fishermen on their boats, NOAA Fisheries has developed solutions to some of the top bycatch challenges facing our nation's fisheries.

Ongoing regional projects include:

- Development of radio and satellite smart buoys to help fishermen avoid bycatch hot spots.
- Understanding sea turtle entanglement in vertical trawls.
- Evaluating the use of elevated sweeps on a West Coast groundfish bottom trawl to reduce contact with the seabed and the corals living there.

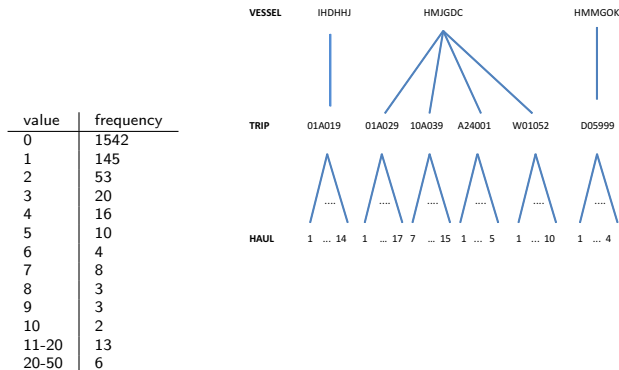
Related Links

- Bycatch Reduction Engineering Program (BREP)
- 2018 BREP Awards
- National Bycatch Program
- Bycatch Science

River herring is one of the species on which BREP-funded projects will focus this year.



Data description: hammerhead sharks



1825 observations on 292 different trips (1 to 21 hauls per trip).

Bycatch counts: 85% zeros, positive counts from 1 to 46.

Covariates: year, average hook depth, area (South Atlantic Bight or Mid Atlantic Bight), season and catch effort (=logarithm of the number of hooks).

Goals and challenges

- Model the data.
: account for all features, dependencies and sources of variability.
- Fit the model.
: satisfactorily approximate the likelihood.
- Optimally predict at the observation level.
: derive the best predictors and estimate them.
- Estimate the variance or MSEP of the predictions.
: derive the MSEP and estimate it.

🌀 Hurdle model with random effects

Responses y_{ij} ($i = 1, \dots, c$ clusters, $j = 1, \dots, n_i$ obs. per cluster).

Independence between clusters. Covariates $\mathbf{x}_{ij}$ and $\mathbf{z}_{ij}$.

Random effects vectors $\mathbf{u}_i$, with $\mathbf{u}_i \sim g_i(\mathbf{u}_i, \boldsymbol{\tau})$, and $\mathbf{v}_i$, with $\mathbf{v}_i \sim h_i(\mathbf{v}_i, \boldsymbol{\kappa})$. Conditional density of y_{ij} :

$$[y_{ij} \mid \mathbf{u}_i, \mathbf{v}_i] = \begin{cases} 1 - p(\mathbf{x}_{ij}, \mathbf{u}_i) & y_{ij} = 0 \\ p(\mathbf{x}_{ij}, \mathbf{u}_i) f(y_{ij}, \lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)) & y_{ij} = 1, 2, 3, \dots \end{cases}$$

where $f(y_{ij}, \lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i))$ is a discrete distribution on the positive.

$$\text{logit}\{p(\mathbf{x}_{ij}, \mathbf{u}_i)\} = \mathbf{x}_{ij}^T \boldsymbol{\alpha} + \sum_{k=1}^a \sigma_{uk} \mathbf{w}_{ijk}^T \mathbf{u}_{ik} \quad (\sigma_u u_i)$$

$$\log\{\lambda(\mathbf{z}_{ij}, \mathbf{u}_i, \mathbf{v}_i)\} = \mathbf{z}_{ij}^T \boldsymbol{\beta} +$$

$$\sum_{k=1}^{b_r} \mathbf{r}_{ijk}^T \{\gamma_k \sigma_{uk} \mathbf{u}_{ik} + \sigma_{rk} \mathbf{v}_{rik}\} + \sum_{k=1}^{b_s} \sigma_{sk} \mathbf{s}_{ijk}^T \mathbf{v}_{sik} \quad (\gamma \sigma_u u_i + \sigma_v v_i.)$$



The marginal likelihood

For $\theta^T = (\alpha, \sigma_{u1}, \dots, \sigma_{ua}, \beta, \gamma, \sigma_{r1}, \dots, \sigma_{rb_r}, \sigma_{s1}, \dots, \sigma_{sb_s}, \tau, \kappa)$, the marginal likelihood writes

$$\begin{aligned} L(\theta) = \prod_{i=1}^c [\mathbf{y}_i] &= \prod_{i=1}^c \int \int [\mathbf{y}_i | \mathbf{u}_i, \mathbf{v}_i] g_i(\mathbf{u}_i, \tau) h_i(\mathbf{v}_i, \kappa) d\mathbf{u}_i d\mathbf{v}_i. \\ &= \prod_{i=1}^c \int \int \prod_{j=1}^{n_i} [y_{ij} | \mathbf{u}_i, \mathbf{v}_i] g_i(\mathbf{u}_i, \tau) h_i(\mathbf{v}_i, \kappa) d\mathbf{u}_i d\mathbf{v}_i. \end{aligned}$$

Likelihood computed numerically by simple Monte Carlo.
Numerical optimization of the likelihood with `optim` in R.
Numerical Hessian matrix also obtained from this function.



Targets and mean squared error

The targets are all of the form $t(\mathbf{u}_i, \mathbf{v}_i, \mathbf{x}, \mathbf{z}, \boldsymbol{\theta})$. The mean squared error of prediction writes:

$$\begin{aligned} \text{mse}_{p_i}(\hat{T}_t, t) &= E_{\mathbf{u}_i, \mathbf{v}_i, \mathbf{y}_i} \left(T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \hat{\boldsymbol{\theta}}; \mathbf{y}_i) - T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \boldsymbol{\theta}; \mathbf{y}_i) \right)^2 \\ &\quad + E_{\mathbf{u}_i, \mathbf{v}_i, \mathbf{y}_i} \left(T_t(\mathbf{x}_{ij}, \mathbf{z}_{ij}, \boldsymbol{\theta}; \mathbf{y}_i) - t(\mathbf{u}_i, \mathbf{v}_i, \mathbf{x}_{ij}, \mathbf{z}_{ij}, \boldsymbol{\theta}) \right)^2 \end{aligned}$$

The best predictors (minimizing mse_p) are given by

$$\begin{aligned} T_t(\mathbf{x}, \mathbf{z}, \boldsymbol{\theta}; \mathbf{y}_i) &= \int \int t(\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{z}, \boldsymbol{\theta}) [\mathbf{u}, \mathbf{v} | \mathbf{y}_i] d\mathbf{u} d\mathbf{v} \\ &= \frac{\int \int t(\mathbf{u}, \mathbf{v}, \mathbf{x}, \mathbf{z}, \boldsymbol{\theta}) [\mathbf{y}_i | \mathbf{u}, \mathbf{v}] g(\mathbf{u}, \boldsymbol{\tau}) h(\mathbf{v}, \boldsymbol{\kappa}) d\mathbf{u} d\mathbf{v}}{\int \int [\mathbf{y}_i | \mathbf{u}, \mathbf{v}] g(\mathbf{u}, \boldsymbol{\tau}) h(\mathbf{v}, \boldsymbol{\kappa}) d\mathbf{u} d\mathbf{v}} \end{aligned}$$

Simple Monte Carlo to obtain empirical best predictors

$\hat{T}_t(\mathbf{x}, \mathbf{z}, \hat{\boldsymbol{\theta}}; \mathbf{y}_i)$ ($\boldsymbol{\theta}$ is replaced by $\hat{\boldsymbol{\theta}}$).



Estimation of msep by parametric bootstrap

Estimate $\text{msep}(\hat{T}_t, t)$ with the following algorithm:

- Compute the estimate $\hat{\theta}$ from the data.
- For $b = 1, \dots, B$
 - use the bootstrap to generate $\hat{\theta}_b^*$ (fast bootstrap, in fact)
 - generate $\mathbf{u}_{b1}^*, \mathbf{u}_{b0}^*$ from $g(\mathbf{u}, \hat{\tau})$, $\mathbf{v}_{b1}^*, \mathbf{v}_{b0}^*$ from $h(\mathbf{v}, \hat{\kappa})$ and y_{bi}^* from the density $[y_i | \mathbf{u}_{b1}^*, \mathbf{v}_{b1}^*]$.
- Compute the bootstrap estimate of $\text{msep}_i(\hat{T}_t, t)$ as

$$\frac{1}{B} \sum_{b=1}^B \left[\hat{T}_t(\mathbf{x}_i, \mathbf{z}_i, \hat{\theta}_b^*, y_{bi}^*) - t(\mathbf{u}_{b0}^*, \mathbf{v}_{b0}^*, \mathbf{x}_i, \mathbf{z}_i, \hat{\theta}) \right]^2.$$

Simulation

Evaluate our **dependent** hurdle mixed effects model (with correlated random effects) and contrast it to the **independent** hurdle model which fits the two parts of the hurdle model separately.

truncated Poisson hurdle model with random intercepts
 $u_{i1} \sim N(0, 1)$ and $v_{i1} \sim N(0, 1)$.

$c = 100$ clusters (half with 5 and half with 10 measurements).

$\mathbf{x}_{ij} = (1, N(0, 1), \text{Bernoulli}(1/2))$ and

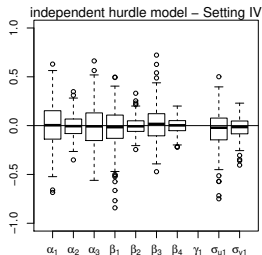
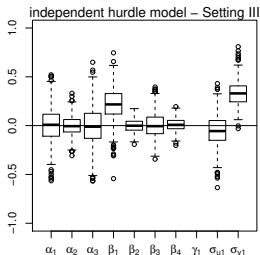
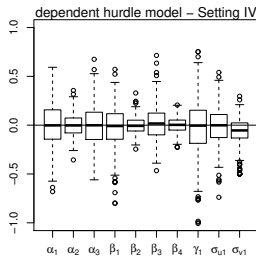
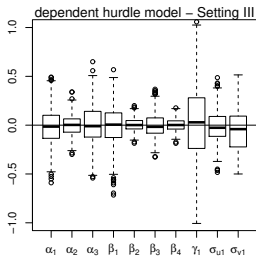
$\mathbf{z}_{ij} = (1, N(0, 1), \text{Bernoulli}(1/2), N(0, 1))$.

400 simulations, $K = 1000$ (likelihood approximation)

$B = 1000$ bootstraps.

Four parameters settings (proportion of zeros of either 0.3 or 0.75 on average, dependent or independent hurdle generating model).

Parameter estimates - Settings 3 and 4



Simulation messages

Parameter estimation:

- For the **dependent** hurdle model:
 - All the regression parameters are estimated unbiasedly in the four settings. The dependence parameter is also approximately unbiased, but with large variability. The “variance components” are slightly underestimated.
 - The confidence intervals for the regression parameters meet the nominal coverage, even more so for the independent setting. The results for the parameters of the random components are less satisfactory.
- For the **independent** hurdle model:
 - some of the regression parameters are biased when the data generating process is dependent (in particular, in the abundance part of the model).
 - Due to software restriction (SE availability), confidence interval are available only for the regression parameters. They perform ok.

Simulation messages

Prediction, for the **dependent** hurdle model:

- Small bias, more often negative, due to the underestimation of the spread parameters.
- Agreement between average estimated msep and empirical msep through simulation (with some exceptions).
- Normal prediction intervals for random intercepts show good coverage for u_i (presence/absence equation), but not so good for v_i (abundance equation).

In summary: ignoring the dependence can lead to bias in the fixed-effects regression parameters as well as an inability to predict random effects in some cases.

Model for hammerhead sharks

Truncated Poisson hurdle model with random intercepts

$u_{i1} \sim N(0, 1)$ and $v_{i1} \sim N(0, 1)$.

$K = 5000$ variates for the likelihood approximation.

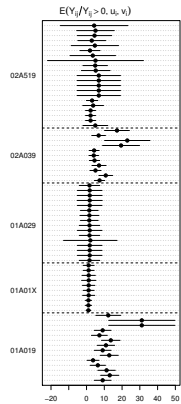
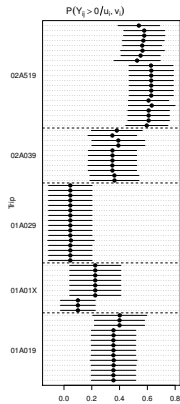
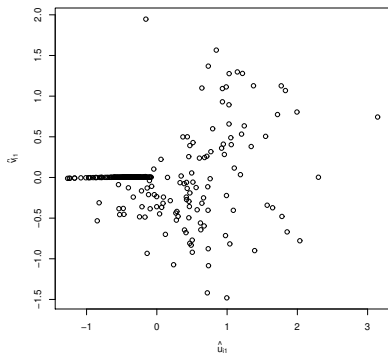
30 starting points for the likelihood maximization (package `fields` in R).

$B = 1000$ bootstrap samples.

Parameter estimates and standard errors

Variable	Presence-absence dep. hurdle model	indep. hurdle model
	Coeff. (SE-Hess, SE-boot)	Coeff. (SE)
Intercept	-2.123 (1.453,1.636)	1.951 (1.508)
YEAR	-0.059 (0.028,0.027)	-0.044 (0.030)
AVGHKDEP	-0.011 (0.011,0.016)	0.007 (0.010)
AREA5	-0.241 (0.242,0.317)	-0.053 (0.254)
SEASONspring	1.609 (0.341,0.351)	1.630 (0.358)
SEASONsummer	0.074 (0.362,0.372)	0.096 (0.366)
SEASONwinter	1.068 (0.369,0.342)	0.950 (0.393)
log(TOTHOOK)	-0.008 (0.212,0.206)	-0.170 (0.222)
Abundance		
Intercept	-4.871 (0.661,0.678)	-3.322 (1.148)
YEAR	-0.132 (0.032,0.029)	-0.105 (0.042)
AVGHKDEP	-0.067 (0.008,0.013)	-0.052 (0.013)
AREA5	-0.151 (0.230,0.257)	-0.182 (0.249)
SEASONspring	0.850 (0.362,0.236)	-0.121 (0.519)
SEASONsummer	-0.435 (0.488,0.401)	-0.843 (0.590)
SEASONwinter	1.278 (0.342,0.226)	0.288 (0.574)
log(TOTHOOK)	1.020 (0.096,0.122)	0.944 (0.171)
γ_1	1.413 (0.159,0.157)	—
σ_{u1}	1.248 (0.145,0.134)	1.387 (n.a.)
σ_{v1}	1.116 (0.144,0.179)	1.544 (n.a.)

Prediction of the targets



*E. Cantoni, J. Mills Flemming and Alan H. Welsh (2017)
A random-effects hurdle model for predicting bycatch of
endangered marine species, Annals of Applied Statistics,
11 (4), 2178-2199. DOI: 10.1214/17-AOAS1074*